Statistics Basics

Q1. What is the difference between descriptive statistics and inferential statistics? Explain with examples

Answer:-> Difference between descriptive and inferential statistics:

Definition:

Descriptive Statistics:

When the whole of the data is used as a sample, the type of sampling is known as Descriptive statistics.

Inferential Statistics:

When not the whole of the data but some portion of the data is used as a sample, the type of sampling is known as inferential statistics.

Use-case:

Descriptive Statistics:

In this type of statistics, it does’nt matters the size of the population, we take the whole population into consideration and compute the result.

Inferential Statistics:

In this type of statistics, the size of the population is taken into account, and that’s why a sample of the population is taken to compute the result.

Example:

Descriptive Statistics:

An example of descriptive Statistics can be computing the average height of students enrolled in Data Science in a batch of 50 students.

Inferential Statistics:

An example of inferential statistics can be computing the average height of all the students whose major is Data Science in the whole world.

So, this was all about it.

Q2. What is sampling in statistics? Explain the differences between random and stratified sampling.

Answer:-> Sampling in statistics:

Sampling in statistics is a method of selecting sample from a particular data for computations.

There are multiple types of sampling used, these are:

1. Simple Random Sampling
2. Stratified Sampling
3. Cluster Sampling
4. Systematic Sampling

The differences between random and stratified sampling:

Definition:

Simple Random Sampling:

Simple Random Sampling is the one in which from a data, some data is selected at random.

There are chances that in Simple Random Sampling that no sample is selected.

Stratified Sampling:

Stratified sampling is the one in which groups are being made and out of all those groups some data is being selected such that every group contributes in the sample in form or the other.

Ideal:

Simple Random Sampling:

Simple Random Sampling is not more accurate because simple random sampling may miss some samples that can be more accurate for that particular case.

Stratified Sampling:

Stratified sampling is more accurate because stratified sampling involves samples from each group being made such that if samples from some groups are more necessary for that case then in that case it performs well.

So, this was all about it.

Q3. Define mean, median, and mode. Explain why these measures of central tendency are important.

Answer:-> Measures of central tendency are of three types:

Mean
Median
Mode

Let us understand them in detail:

1. Mean: Mean is defined as the arithmetic mid point of the data. Mean is computed as the summation of all the observations in the data to the total number of observations in the data. Mean is the only central tendency that is affected by the presence of outlier in the data.

2. Median: Median is defined as the physical mid point of the data. Median is computed by two ways one for the odd number of observations and one for the even number of observations.

When the number of observations are even, we remove every observation from both the sides till there are two middlemost observations left and then we sum up those two observations and divide them by 2 to get the median of the data.

When the number of observations are odd, we simply do the same to get that one number that is left in the middle.

3. Mode: Mode is defined as the observation which occur the most number of times in a particular data.

The observation with highest number of frequency is mode in a data.

Measures of central tendency are important because they represent the data by the help of a single value.

Also, the measures of central tendency help whenever there is an outlier or null value, so in the replacement of those particular values, the measures of central tendency are used.

So, this was all about it.

Q4. Explain skewness and kurtosis. What does a positive skew imply about the data?

Answer:-> Skewness and kurtosis:

Skewness tells us about the shape of the data about which side the data is aligned towards, giving us information about the symmetricity of the data.

There are three types of skewness:

1. Zero symmetricity: Zero symmetricity is the case where Mean=Median=Mode and also the Quartile 2 is exactly in the middle of Quartile 1 and Quartile 3.

2. Positively skewed: A positively skewed data is the one in which the data is aligned to the right side.

3. Negatively skewed: A negatively skewed data is the one in which the data is aligned to the left side.

A positive skew imply about the data that the data is aligned towards the right side.

So, this was all about it.

Q5. Implement a Python program to compute the mean, median, and mode of a given list of numbers.

numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28].

Answer:-> Python code is as follows:

```
import numpy as np
from scipy import stats
```

```
numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]

print("The mean of the numbers data is :-> ", np.mean(numbers))
print("The median of the numbers data is :-> ", np.median(numbers))
print("The mode of the numbers data is :-> ", stats.mode(numbers))
```

```
The mean of the numbers data is :->  19.6
The median of the numbers data is :->  19.0
The mode of the numbers data is :->  ModeResult(mode=np.int64(12), count=np.int64(3))
```

Q6. Compute the covariance and correlation coefficient between the following two datasets provided as lists in Python:

list_x = [10, 20, 30, 40, 50]
list_y = [15, 25, 35, 45, 60]

Answer:-> Pythons code is as follows:

```
list_x = [10, 20, 30, 40, 50]
list_y = [15, 25, 35, 45, 60]

df = pd.DataFrame({
    'list_x': list_x,
    'list_y': list_y
})

print("The covarince of the data is :-> ", df.cov())
print("The correlation of the data is :-> ", df.corr())
```

```
The covarince of the data is :->         list_x  list_y
list_x   250.0   275.0
list_y   275.0   305.0
The correlation of the data is :->           list_x    list_y
list_x  1.000000  0.995893
list_y  0.995893  1.000000
```

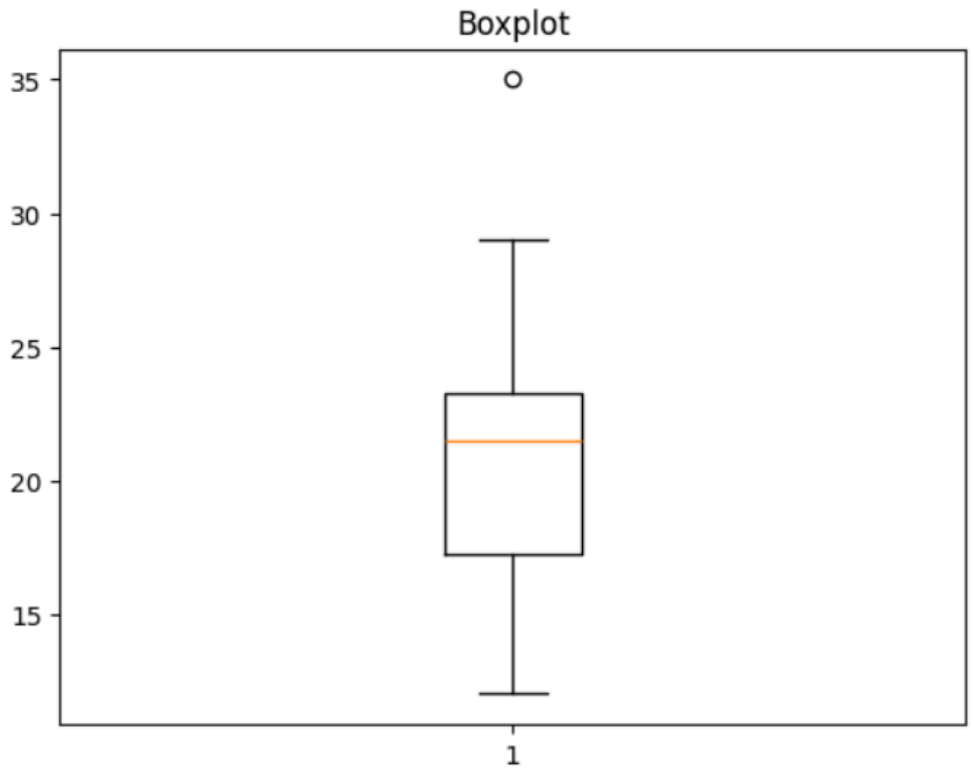
Q7. Write a Python script to draw a boxplot for the following numeric list and identify its outliers. Explain the result:

data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]

Answer:-> Python code is as follows:

```
import matplotlib.pyplot as plt

data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]

plt.boxplot(data)
plt.title("Boxplot")
plt.show()

Q1 = np.percentile(data, 25)
Q2 = np.percentile(data, 50)
Q3 = np.percentile(data, 75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = [x for x in data if x < lower_bound or x > upper_bound]

print("Q1:", Q1)
print("Median:", Q2)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
print("Outliers:", outliers)
```

...




```
Q1: 17.25
Median: 21.5
Q3: 23.25
IQR: 6.0
Lower Bound: 8.25
Upper Bound: 32.25
Outliers: [35]
```

Q8. Question 8: You are working as a data analyst in an e-commerce company. The marketing team wants to know if there is a relationship between advertising spend and daily sales.

● Explain how you would use covariance and correlation to explore this relationship.
● Write Python code to compute the correlation between the two lists:

advertising_spend = [200, 250, 300, 400, 500]
daily_sales = [2200, 2450, 2750, 3200, 4000]

Answer:-> As a data analyst, covariance and correlation are useful to explore this relationship as covariance tells us about the direction of the relationship.

Covariance is used to tell that whether the relation between the two variables is positive or negative. It isn't affected by the magnitude as it is a dimensioned quantity.

Correlation is used to find the range in which it's relationship between two variables lies. It is a dimension less quantity. It is usually lies in the range of -1 to 1.

Python code for correlation between two variables is as follows:

```
advertising_spend = [200, 250, 300, 400, 500]
daily_sales = [2200, 2450, 2750, 3200, 4000]

df = pd.DataFrame({
    'Advertising Spend': advertising_spend,
    'Daily Sales': daily_sales
})

print("The correlation is :-> ", df.corr())
```

```
The correlation is :->                    Advertising Spend  Daily Sales
Advertising Spend           1.000000     0.993582
Daily Sales                 0.993582     1.000000
```

Q9. Your team has collected customer satisfaction survey data on a scale of 1-10 and wants to understand its distribution before launching a new product.

- Explain which summary statistics and visualizations (e.g. mean, standard deviation, histogram) you'd use.
- Write Python code to create a histogram using Matplotlib for the survey data:

survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]

Answer:-> For this customer satisfaction survey data, i will be using the Measures of central tendency, also standard deviation and range for the same purpose.

The reason for using these are as follows:

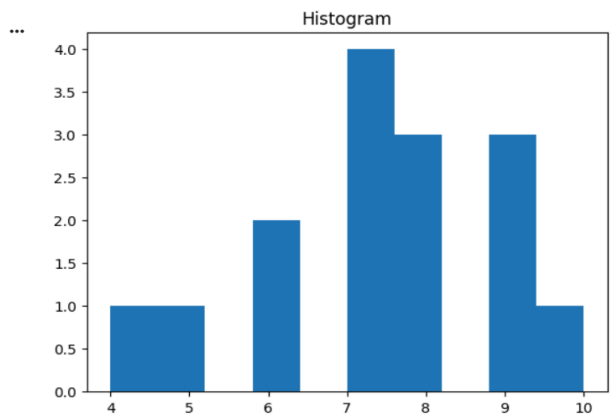
1. Mean: Will help us in understanding what is the mean rating the survey has generated on that product.

2. Median: Will help us in understanding that if there is any outlier rating, then what is the actual median rating around which all the ratings are settled.

3. Mode: Will help us in understanding which is that one rating which is the most frequent.

4. Range: Will help us in understanding the range in which the rating lies in.

5. Standard Deviation: Will help us in understanding the spread of the data.

Python code is as follows:

```
survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]

plt.hist(survey_scores)
plt.title("Histogram")
plt.show()
```